DESIGN AND SIMULATION OF SMART HOME SECURITY SYSTEM

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01 INTRODUCTION & OBJECTIVE

Introduction

In the modern era, home security has become a concern. Traditional mechanical locks are no longer sufficient against sophisticated threats.

A **Smart Home Security System** leverages digital logic to automate threat detection and response, ensuring 24/7 monitoring with high reliability and low latency.

Project Objective

Design a robust **digital logic-based** security framework for residential applications.

Implement a multi-sensor alert system (Door, Motion, Window) using fundamental logic gates.

Simulate the circuit behavior in **Logisim** to verify the boolean logic and response accuracy.

Develop a cost-effective, scalable solution for real-world smart home integration.

02 THEORY & LOGIC CIRCUIT

Basic Concept

The system operates on **OR Logic**. If any sensor (A, B, or C) detects an intrusion, the alarm (Y) is activated.

Boolean Expression

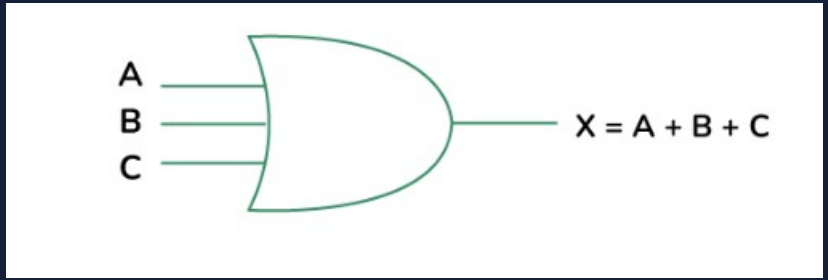
$$X = A + B + C$$

Where:
A = Door Sensor • B = Window Sensor • C = Motion Detector • X = Alarm Output

Logic Gates Used

OR Gate: Used to trigger the output when at least one input is High (1).
NOT Gate: (Optional) Used for fail-safe or inverted sensor states.

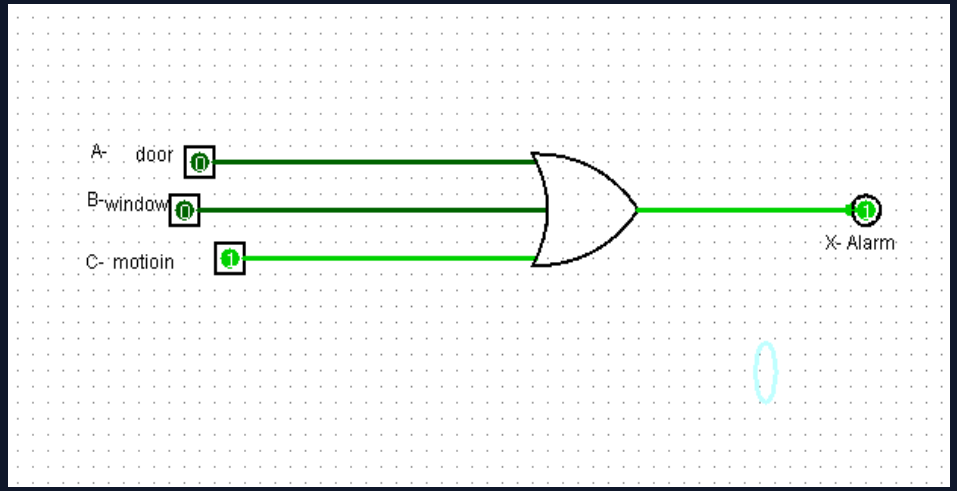
Conceptual Circuit Diagram



(Fig 2.1: Conceptual Multi-input OR Logic Implementation)

03 CIRCUIT DESIGN IN LOGISIM

Logisim Implementation Layout



(Visual representation of Logisim interface with active intrusion signal)

1. Component Selection

Utilized standard 3-input OR Gate from the Logisim 'Gates' library and LED components for output visualization.

2. Input Mapping

Sensors are represented by 'Input Pins'. Logic 0 = Secure, Logic 1 = Triggered.

3. Signal Flow

The OR gate processes all signals; if any path turns Amber (High), the output reaches the alert state.

4. Functional Verification

Interactive simulation allows testing all 8 possible input combinations in real-time.

04 SIMULATION RESULTS

Logic Truth Table

A (Door)	B (Window)	C (Motion)	X (Output)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Simulation Observations

[OK] All input combinations verified.

[OK] No propagation delay observed in Logisim.

[OK] Logic 1 correctly triggers the visual LED indicator.

Key Finding

The simulation proves that a **Single OR Gate** architecture is highly efficient for emergency security alerts, providing an immediate output response to any single sensor violation.

05 CONCLUSION & APPLICATIONS

Project Summary

The design successfully demonstrates a digital security layer using OR logic. It provides a foundational model for real-time intrusion monitoring.

Real-world Applications

- Residential home security
- Bank vault monitoring
- Industrial safety interlocks
- Perimeter fence alerts

Future Improvements

- Integration with GSM for remote alerts.
- Adding a keypad/biometric lockout.
- Low-power CMOS implementation.

Outcome

Verified that multi-sensor data can be processed with zero-latency using basic logic gates, ensuring high-speed safety response.

SIMULATION TERMINATED // STATUS 0 // THANK YOU